Clustering Overview

Clustering algorithms are a **type of unsupervised learning algorithms** where we have a dataset containing **columns (features)** but **no labels attached to them.**

In supervised learning we are given labels, but in clustering **these labels have to be discovered or provided by us** after the algorithm groups similar data points.

The goal of clustering is to **divide the dataset into groups (clusters)** such that:

- Data points **within the same cluster are similar**
- Data points **in different clusters are dissimilar**

Example Scenario

Imagine a dataset containing:

- Customer **spending values**
- Customer **retention time**

Our goal is to classify customers into groups such as:

Spending	Retention	Cluster Example
High	Long	Loyal High-Value Customers
High	Short	High Spenders with Low Loyalty
Low	Long	Budget Loyal Customers
Low	Short	Low Value Customers

Since the dataset does **not contain labels**, we must **discover these groups using a clustering algorithm**, and afterwards we can **assign meaningful labels to the clusters ourselves.**

Visual Intuition

Clusters can be visualized as **groups of nearby points in feature space.**

Points that are **close together form clusters**, while distant points belong to different clusters.

```
Retention
↑
| ● ● ●
| ● ● ● ● Cluster 1
|
| ● ● ●
| ● ● ● Cluster 2
|
| ● ●
| ● ● ● Cluster 3
|
+──────────────────────────────→ Spending
```

Types of Clustering

Clustering can broadly be categorized into two types.

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1) Hard Clustering

In **hard clustering**, each data point belongs to **exactly one cluster.**

Clusters **do not overlap**, meaning a datapoint cannot simultaneously belong to multiple clusters.

Hard Clustering Visualization

```
Cluster A Cluster B

● ● ● ● ● ● ●
● ● ● ● ●
● ● ●
```

A datapoint appears **in only one cluster region.**

Example

If we classify customer data:

- A **high spending + long retention customer** cannot simultaneously be classified as
- **low spending + short retention**

Each datapoint is assigned **one and only one cluster.**

Example Algorithms

Algorithm	Description
K-Means	Most common centroid-based clustering algorithm
K-Medoids	Similar to K-Means but uses actual datapoints as centers
Hierarchical Clustering	Builds clusters in a tree-like structure

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2) Soft Clustering

In **soft clustering**, clusters **can overlap.**

Instead of assigning a datapoint to a single cluster, the algorithm assigns a **probability or degree of membership** to multiple clusters.

This means a datapoint may **partially belong to multiple clusters.**

Soft Clustering Visualization

```
    Cluster A
   ● ● ● ●
 ●      ●
```

● ● ← Data point partially
● ● belonging to both
● ● ● ●

```
    Cluster B
   ● ● ● ●
```

The datapoint lies in the **overlapping region** of clusters.

Example

Customer	Cluster A	Cluster B
C1	0.8	0.2
C2	0.4	0.6
C3	0.2	0.8

This means:

- Customer C1 belongs **80% to Cluster A**
- Customer C1 belongs **20% to Cluster B**

Example Algorithms

Algorithm	Description
Gaussian Mixture Models (GMM)	Uses probability distributions
Fuzzy C-Means	Assigns fuzzy membership values

Different Methods to Form Clusters

Clustering algorithms can also be categorized based on **how they form clusters.**

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1) Centroid-Based Clustering

Centroid-based clustering forms clusters around **central points called centroids.**

A **centroid** represents the **mean position of all datapoints within a cluster.**

The most common algorithm in this category is **K-Means.**

Working Principle

1. Choose the number of clusters **K**
2. Randomly initialize **K centroids**
3. Assign each datapoint to the **nearest centroid**
4. Recalculate centroids using the mean of assigned points
5. Repeat until cluster assignments stabilize

Centroid Visualization

Cluster

```
  • •
```

• • •

```
    X   ← Centroid
```

• • •
• •

`X` represents the **centroid (mean position of cluster points).**

Advantages

- Simple and easy to implement
- Efficient for large datasets
- Works well when clusters are roughly spherical

Limitations

- Requires choosing **K beforehand**
- Sensitive to **initial centroid placement**
- Performs poorly with **irregular cluster shapes**

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2) Hierarchical Clustering

Hierarchical clustering builds clusters in a **tree-like structure called a dendrogram.**

There are two main approaches.

Method	Description
Agglomerative	Bottom-up approach where small clusters merge together
Divisive	Top-down approach where large clusters are split

Agglomerative Clustering Process

In **agglomerative clustering**:

1. Start with **each datapoint as its own cluster**
2. Find the **two closest clusters**
3. Merge them into a **single cluster**
4. Repeat until **all points belong to one cluster**

Agglomerative Visualization

Step 1: Each point is its own cluster

A B C D

Step 2: Closest points merge

(A,B) C D

Step 3: Next closest merge

(A,B) (C,D)

Step 4: Final merge

((A,B),(C,D))

Dendrogram Representation

The hierarchy of merges can be represented using a **dendrogram.**

```
    _______ A
   |
__| B
| |
| |__ C
|
|____ D
```

Interpretation

- The **height where branches merge** represents the **distance between clusters.**
- Cutting the dendrogram at different levels gives **different numbers of clusters.**

Example:

Cut here → 2 clusters

```
    _______ A
   |
__| B
|
|__ C
|
|____ D
```

Clusters:

- Cluster 1 → {A, B}
- Cluster 2 → {C, D}

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3) Density-Based Clustering

Density-based clustering forms clusters based on **regions with high density of datapoints.**

Areas with **many nearby points form clusters**, while sparse regions are considered **noise or outliers.**

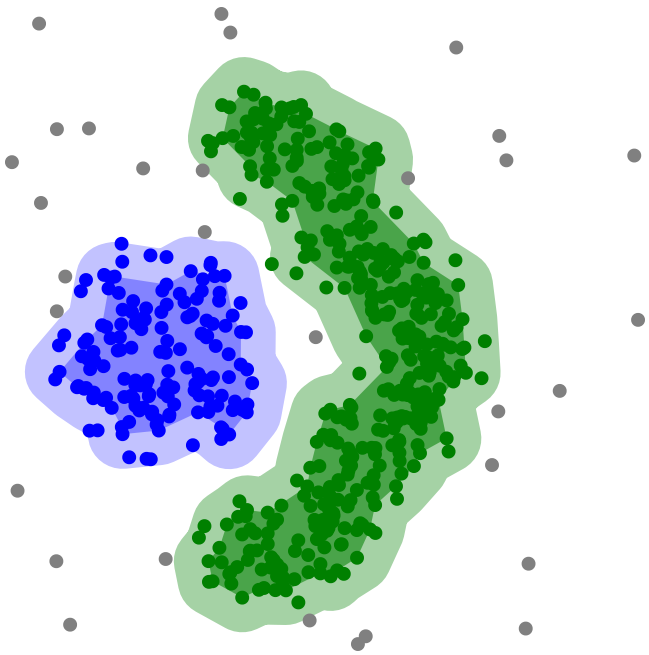
Example Algorithm

DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

Advantages

- Detects **arbitrarily shaped clusters**
- Handles **noise and outliers well**

Visualization



Summary

Clustering Type	Key Idea	Example Algorithms
Hard Clustering	Each datapoint belongs to exactly one cluster	K-Means
Soft Clustering	Datapoints can belong to multiple clusters	Gaussian Mixture Models
Centroid-Based	Clusters formed around centroids	K-Means

Clustering Type	Key Idea	Example Algorithms
Hierarchical	Clusters built as a tree structure	Agglomerative Clustering
Density-Based	Clusters formed based on dense regions	DBSCAN